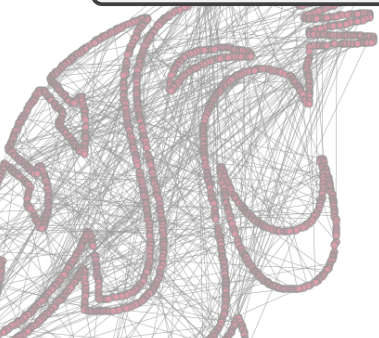


A Science Chit Chat by a Mathematician

Garrett Kepler

April 8th, 2026



Outline

① Probability Theory

② Gerrymandering

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② Gerrymandering



Weak Law of Large Numbers (WLLN)

Intuitively

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Complicatedly

If $X_1, X_2, \dots$ are a sequence of independent identically distributed random variables with $\mathbb{E}[X] < \infty$, then for any $\epsilon > 0$

$$\lim_{n \rightarrow \infty} \mathbb{P} \left[\left| \frac{1}{n} \sum_{i=1}^n X_i - \mathbb{E}[X_i] \right| < \epsilon \right] = 1$$

WLLN Through Experience

A little intuition builder:

<https://garrett-kepler.github.io/resources/LLN/>

A New School of Thought



Figure: (Left) Pavel Nekrasov: firm believer WLLN only applies to independent events. (Right) Andrey Markov: prover that WLLN applies to dependent events too!

WLLN with Dependence

Intuitively

The beautiful result Markov proved is that we actually don't need independent events! With a little assumptions we can confidently say the average will get close to the expected value.

Complicatedly

If $X_1, X_2, \dots$ is a positive Harris recurrent Markov chain with unique invariant distribution π and f is a real-valued function on the state space, then

$$\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n f(X_i) \xrightarrow{a.s.} \mu = \mathbb{E}_\pi f(X) = \int f(x) \pi(dx)$$

① Probability Theory

② Gerrymandering



History of Gerrymandering

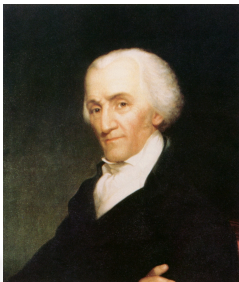


Figure: Eldridge Gerry and his “salamander district” in 1812 Massachusetts: ([Click here for full map](#))

A portrait of John Jay, an older man with white hair, wearing a dark coat and a white cravat. He is looking slightly to the right of the viewer.

Theorem
Gerry + Salamander = Gerrymander

Proof. Proof left to audience as an exercise. □

5

The Maps We See

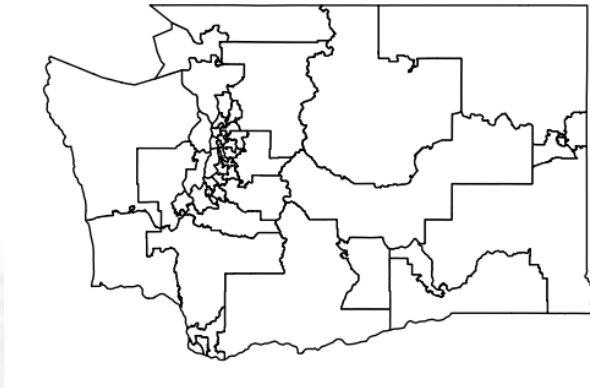


Figure: Washington's 2020 Legislative Districts

The Maps We Deal With

In practice, it is very difficult with maps themselves. So, we simplify them.

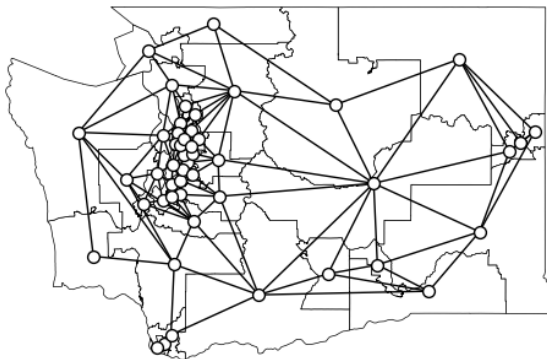


Figure: Washington's 2020 Legislative Districts and corresponding dual graph

The Maps We Deal With

Once we have the dual graphs, we can start thinking about district assignments and metrics of fairness.

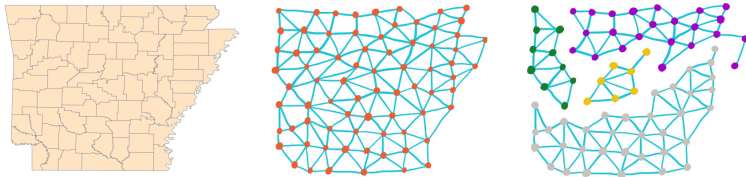


Figure: Example of a dual graph of Arkansas with its district assignment as a partition

Gerrymandering through “Definition”

“Gerrymandering, defined in the contexts of representative electoral systems, is the political manipulation of electoral district boundaries to advantage a party, group, or socioeconomic class within the constituency.” - Wikipedia circa 2:53 PM April 6th, 2026 (For Link Click Here)

Question

Okay, but how do we know when someone is trying to advantage a party/group?

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Question

Okay, but how do we know when someone is trying to advantage a party/group?

Answer (*)

We can define any metric we like!

What Might We Use to Determine Gerrymandering?

There are lots of metrics we use on generated maps:

- Mean-Median Score (score far from zero implies district competitiveness is not balanced)
- Polsby-Popper Score (how compact is your district in comparison to the circle)
- Population Balance (are your districts somewhat balanced in population?)
- Majority-Minority Ratio (how well does your districted map respect minority voting power)
- And more!

And Now We Compare!

Question

Okay, we have a definition of “fairness” we like, how do we determine if a current map is gerrymandered?

Answer (*)

We can draw all the possible maps and compare!

Womp Womp :(

Answer (*)

We could draw all the possible maps and compare to the map someone wants to enact!

Actually, this highly impractical...

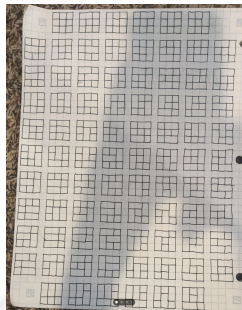
```
In [19]:  start = time.time()
          allperms = list(permutations(partition))
          print('Runtime: ' + str(time.time() - start))

-----
MemoryError                                Traceback (most recent call last)
<ipython-input-19-04af173883b2> in <module>
      1 start = time.time()
----> 2 allperms = list(permutations(partition))
      3 print('Runtime: ' + str(time.time() - start))

MemoryError:
```

Figure: Memory error for just trying to calculate all 4 partitions of grid from game with 16 nodes (circa 5:12 PM April 6th, 2026)

Perspective



... 15 pages later ...

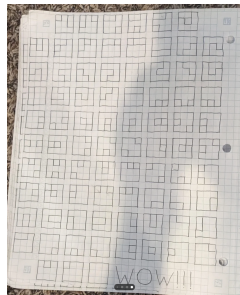


Figure: For perspective, here are the connected partitions of the 3 by 3 grid enumerated by a Reddit user (17 photos total!)

The Recombination Approach

What we do in practice is much more clever:

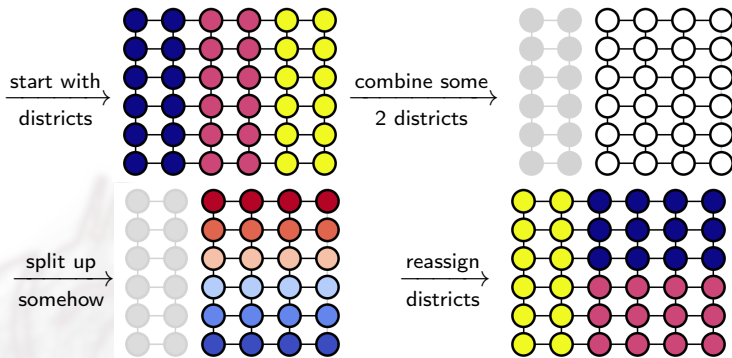
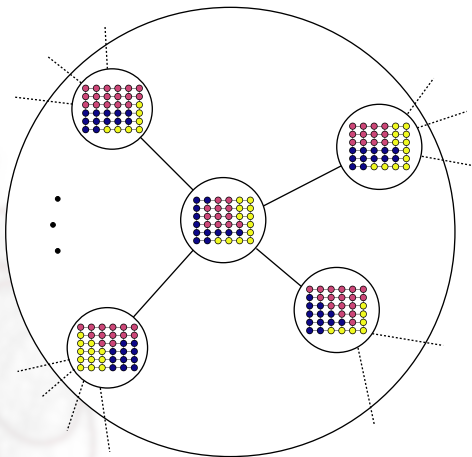


Figure: An example of what we're doing

Metagraph

Now, what we get to do is move around the space of districted maps. Moreover, the way we do it guarantees Markov's LLN applies!



LLN for Maps

Intuitively

If we recombine and split districts cleverly, we can get a *good* set of maps that we can compare to. That way, we can detect gerrymandering when it occurs!

Less Intuitively

Each movement in the districted maps is a dependent yet random sample such that Markov's version of the Law of Large Numbers applies to any metric we use to detect gerrymandering in maps.

Complicatedly

Ideally, we will define an aperiodic Markov chain with one recurrent class such that the Ergodic theorem applies and we have a unique stationary distribution of districted maps. Then, we can get representative samples from the stationary distribution.

Ensemble Analysis

This idea is what we call ensemble analysis: Draw a lot of *relevant* maps and compare them to proposed maps.

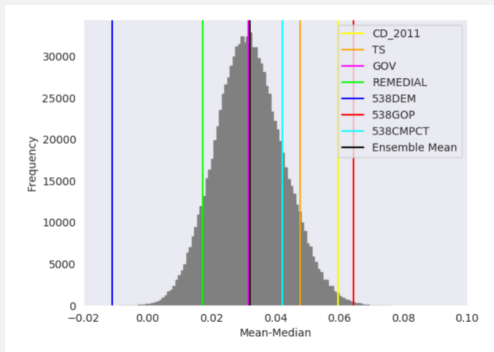


Figure: Comparison of scores from recombination-based maps (grey) with scores of politically proposed maps (vertical lines)

“Fairness”

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So, with all these metrics, when can we say a map is fair?



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Answer

In short, we can't know for certain. We can only carefully tailor Markov chains to provide ample evidence that it might be!

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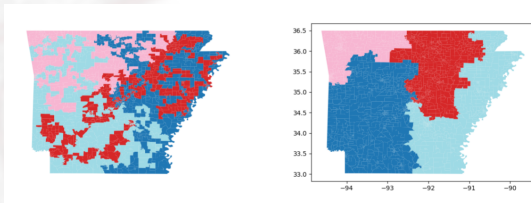
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Answer

In short, we can't know for certain. We can only carefully tailor Markov chains to provide ample evidence that it might be!

While a map may not pass the “eyeball” test, it may be the best option.



“Fairness”

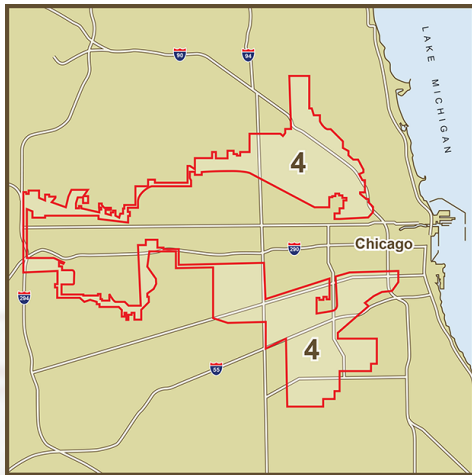
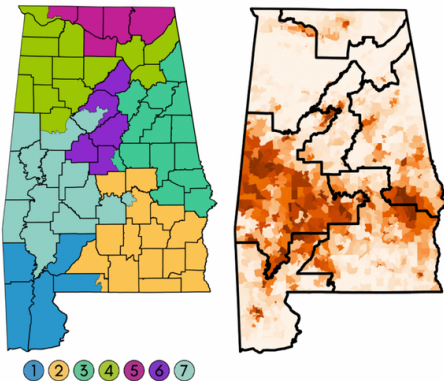


Figure: For example, the famous “Horseshoe District” from Chicago. Making this district score well in compactness would make it score poorly in majority-minority ratio.

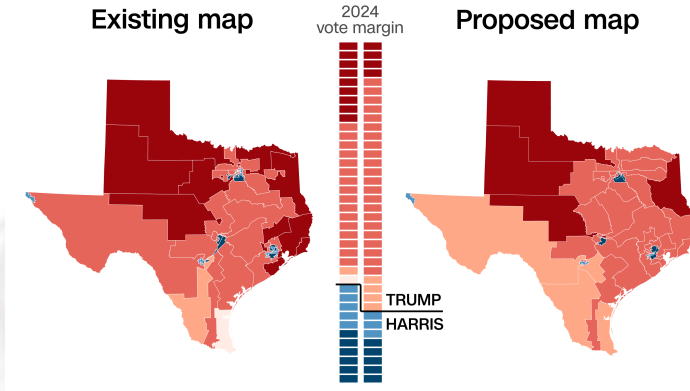
These Questions Still Arise

In 2023, the Supreme Court ruled that Alabama had racially gerrymandered.



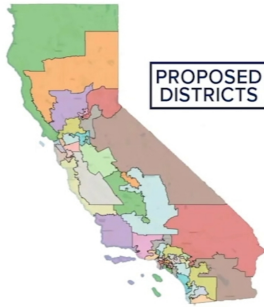
These Questions Still Arise

In 2025, the Supreme Court ruled that Texas had not racially gerrymandered.



These Questions Still Arise

In 2025, California *voted* that they redraw their districted map.



These Questions Still Arise (and There May Be No Answer)

We are not be able to make the perfect map, but we can try and detect bad maps and do better.



Resources

For the curious, here are some redistricting-focused resources:

- “*Political Geometry*” – Textbook on mathematical redistricting methods
<https://data-democracy.org/gerrybook.html>
- “*Redistricting Data Hub*” – Dataset resource with redistricting-oriented data <https://redistrictingdatahub.org/>

Try Gerrymandering Yourself

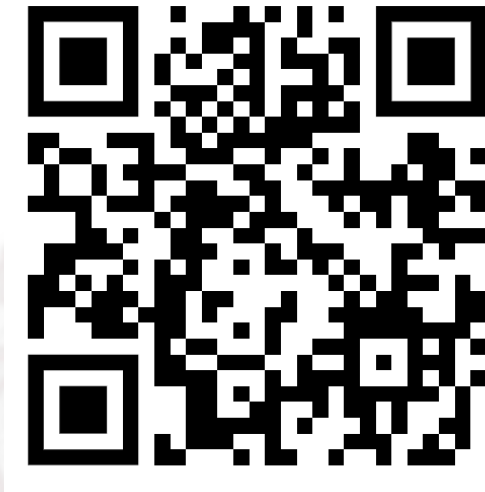


Figure: <https://garrett-kepler.github.io/resources/gerry/>